COMSATS UNIVERSITY ISLAMABAD

LAHORE CAMPUS

ASSIGNMENT 3

Project Planning and Management

Project Title

IntelliSafe: Real-Time Working Site Safety Monitoring using Edge Computer Vision

Name: Muhammad Uzair Qamar

Registration Number: FA23-BCE-098

Table of Contents

1. Introduction . . . 3

2. Work Breakdown Structure (WBS) . . . 3

2.1 Definition and Purpose . . . 3

2.2 WBS Dictionary / Work Package Description . . . 4

2.3 WBS Management Notes . . . 4

3. Organizational Breakdown Structure (OBS) . . . 5

3.1 Definition and Purpose . . . 5

3.2 Roles and Responsibilities . . . 6

3.3 Organizational Notes . . . 6

4. Project Communication Flow . . . 6

4.1 Purpose . . . 6

4.2 How Information Moves . . . 7

4.3 Who Communicates with Whom . . . 7

4.4 Meetings and Updates . . . 8

4.5 When Something Goes Wrong . . . 8

4.6 Ground Rules . . . 9

5. Conclusion . . . 9

1. Introduction

This assignment presents three project-planning components for the Final Year Project (FYP) titled “IntelliSafe: Real-Time Working Site Safety Monitoring using Edge Computer Vision.” The proposed system is intended to monitor industrial and construction sites through a camera connected to an NVIDIA Jetson Nano, detect workers, helmets, vests, fire and smoke with a single YOLOv8n detector, decide per-worker PPE compliance through a compliance engine, and deliver alerts, image evidence and compliance analytics to supervisors through a web dashboard. For fire or smoke, an ESP32 alarm unit sounds a buzzer and strobe on site (target: within one second), even if the internet link is down.

The Work Breakdown Structure (WBS) organizes the project scope into manageable work packages. The Organizational Breakdown Structure (OBS) presents a hypothetical company-style hierarchy for managing and delivering the project. The communication flow defines how instructions, progress updates, technical information, issues, and approvals move between the project team, company leadership, the supervisor, and intended users.

2. Work Breakdown Structure (WBS)

2.1 Definition and Purpose

A Work Breakdown Structure is a hierarchical decomposition of the total project scope into smaller deliverables and work packages. It clarifies what must be produced, supports task assignment and estimation, and helps the team track progress. The WBS below follows the proposed FYP scope and the approved detection scope of five classes: worker, helmet, vest, fire and smoke. Work packages are arranged along the system pipeline, from data and model through edge deployment, messaging, alarm and dashboard, so that the project remains achievable across the two available semesters (FYP-I and FYP-II).

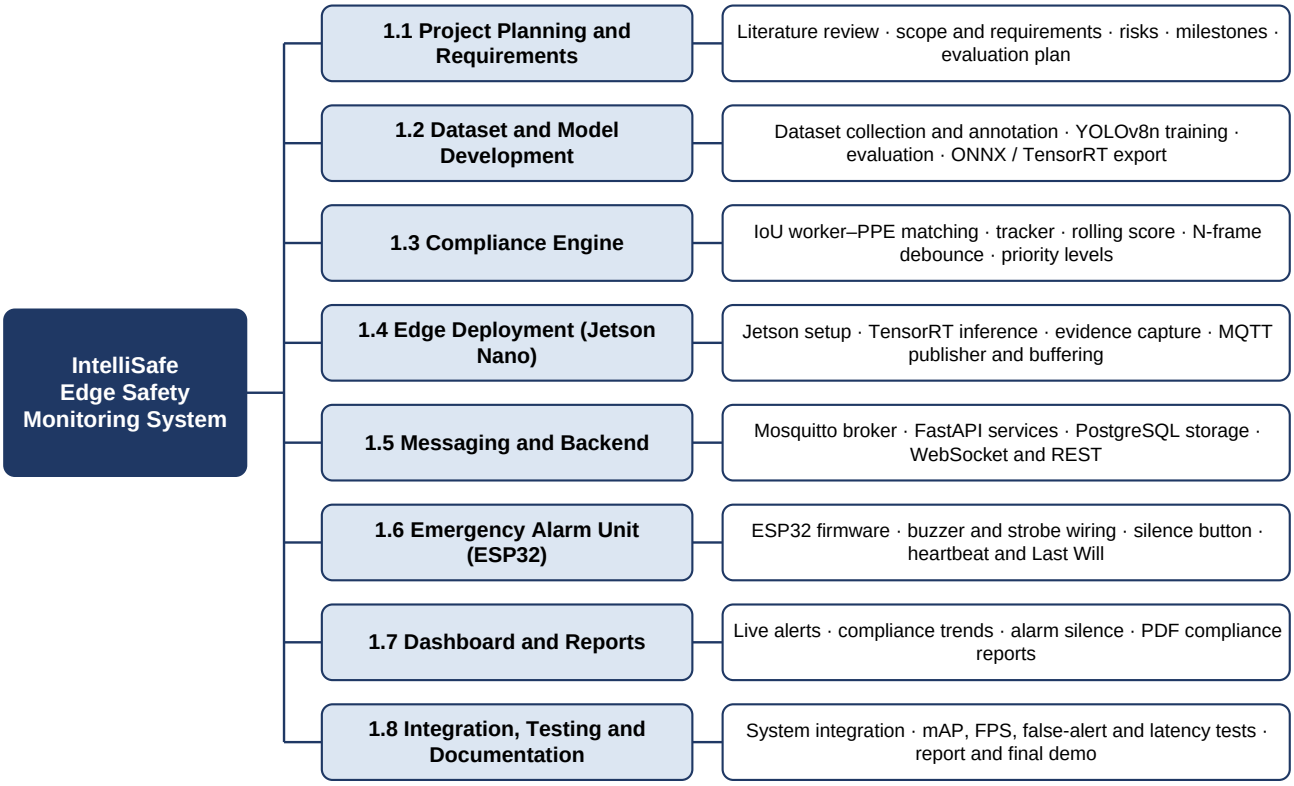

Figure 1. Work Breakdown Structure for the IntelliSafe FYP

2.2 WBS Dictionary / Work Package Description

WBS Code	Work Package	Description / Activities	Expected Deliverable
1.1	Project Planning and Requirements	Review related work (Nath et al., Wang et al., de Venancio et al.); confirm scope, objectives, research questions, constraints, risks, milestones and evaluation plan.	Approved proposal and project plan
1.2	Dataset and Model Development	Collect public datasets (Pictor-v3, CHV, D-Fire) and local frames; annotate in YOLO format; split train/val/test; fine-tune YOLOv8n from COCO weights at 640x640; export .pt to ONNX to TensorRT (FP16).	Annotated dataset and trained five-class YOLOv8n model
1.3	Compliance Engine	Match each helmet and vest to a worker by IoU (> 0.3) with a simple tracker for stable worker_id; compute rolling compliance score; apply N-frame debounce, one alert per worker per violation type; assign priority levels; record evidence.	Working compliance engine with evidence records
1.4	Edge Deployment (Jetson Nano)	Set up Jetson Nano and JetPack; deploy the TensorRT model and engine; capture camera frames; save annotated evidence images; publish JSON events with paho-mqtt (QoS 1) and buffer them if the network drops.	Real-time edge application on the Jetson Nano
1.5	Messaging and Backend	Run the Mosquitto broker on the site LAN; build FastAPI services that receive, validate and store events and evidence in PostgreSQL; provide WebSocket/REST access; use a simulator until the Jetson is ready.	Working backend, database and event pipeline
1.6	Emergency Alarm Unit (ESP32)	Write ESP32 firmware that subscribes to ppe/alarm; wire the active buzzer (NPN driver), red strobe LED and silence button; send a heartbeat every 5 s and an MQTT Last Will.	ESP32 buzzer and strobe alarm unit
1.7	Dashboard and Reports	Build the React dashboard showing live alerts, compliance trends, device status (“alarm offline”) and alarm silence; generate PDF compliance reports.	Supervisor dashboard and PDF reports
1.8	Integration, Testing and Documentation	Integrate edge, backend, alarm and dashboard; run test scenarios; measure mAP@0.5, per-class precision/recall, on-device FPS, false alerts per hour and camera-to-dashboard / camera-to-buzzer delay; prepare FYP report and demo.	Evaluated prototype, documentation and final demonstration

2.3 WBS Management Notes

- The WBS includes the full core scope described in the FYP proposal: five-class detection, per-worker compliance, MQTT event pipeline, evidence storage, on-site ESP32 alarm, dashboard with trends and PDF reports, and on-device evaluation.
- Each work package should be further divided into individual tasks when the team prepares the detailed schedule (the two-semester Gantt chart in the proposal).
- The detection scope is fixed to worker, helmet, vest, fire and smoke. Additional classes or features outside the approved scope should not delay delivery of the core monitoring and alerting system.
- Completion should be assessed against measurable acceptance criteria, including mAP@0.5 of at least 0.70, real-time inference on the Jetson Nano, false alerts per hour with and without debounce, camera-to-buzzer delay of one second or less, and measured camera-to-dashboard delay.

3. Organizational Breakdown Structure (OBS)

3.1 Definition and Purpose

An Organizational Breakdown Structure represents the project organization by showing roles, teams, and reporting relationships. For this assignment, the project is represented as a hypothetical technology company developing an edge-AI industrial safety product. The CEO, co-founders, and managers below are proposed organizational roles for planning purposes; they are not claims that a registered company or these formal job titles already exist. The academic FYP supervisor remains the project's university supervisory stakeholder.

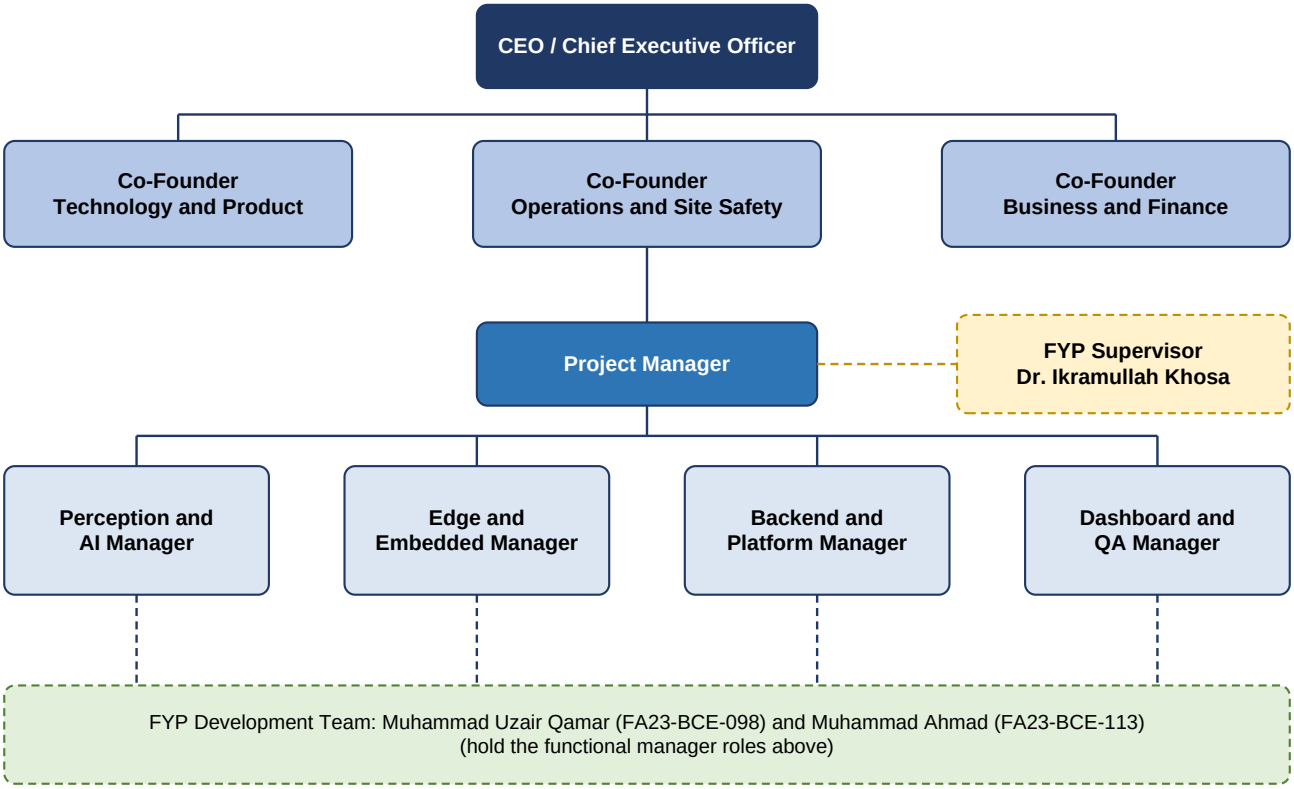

Figure 2. Company-Style Organizational Breakdown Structure (proposed)

3.2 Roles and Responsibilities

Role / Stakeholder	Main Responsibilities
CEO / Chief Executive Officer	Overall strategic direction, project sponsorship, resource priorities, major decisions and high-level review.
Co-Founder – Technology and Product	Oversees technical architecture, integration decisions, product quality and alignment between the edge, backend and dashboard components.
Co-Founder – Operations and Site Safety	Coordinates practical site-safety requirements, supervisor needs, access to work sites and validation context.
Co-Founder – Business and Finance	Tracks budget, equipment procurement (Jetson Nano, camera, ESP32), cost per monitored site, resource use and potential product viability.
Project Manager	Maintains the project plan, schedule, milestones, risk register, meetings, task allocation and progress reporting.
Perception and AI Manager	Leads dataset collection and annotation, YOLOv8n training, evaluation and export, the compliance engine (matching, score, debounce) and evidence capture.
Edge and Embedded Manager	Leads Jetson Nano setup and model deployment, the MQTT publisher with buffering, and the ESP32 alarm unit (firmware and wiring).
Backend and Platform Manager	Leads the MQTT broker, backend services, data validation, PostgreSQL database, evidence storage and APIs.
Dashboard and Quality Assurance Manager	Leads dashboard usability, live alerts and trends, PDF reports, system integration test planning, defect tracking and acceptance evidence.
FYP Supervisor	Dr. Ikramullah Khosa provides academic guidance, reviews progress, advises on technical/research direction and assesses project milestones.
FYP Development Team	Muhammad Uzair Qamar (FA23-BCE-098) and Muhammad Ahmad (FA23-BCE-113) collaboratively develop and evaluate the system. In the proposal, Uzair leads perception and Muhammad Ahmad leads the platform; event format, integration, testing and the final demo are joint. Specific manager assignments can be confirmed by the team.

3.3 Organizational Notes

- The company-style hierarchy is a planning model created to satisfy the assignment's organizational view; it does not imply that a company has been legally formed.
- The CEO and co-founders provide strategic oversight, while the project manager coordinates day-to-day delivery and managers lead functional workstreams.
- The two FYP students form the actual academic project team. They may hold more than one functional role depending on skills and workload.
- The university supervisor is shown as an academic stakeholder rather than an employee reporting to the hypothetical company CEO.

4. Project Communication Flow

4.1 Purpose

Good communication keeps everyone working from the same plan. This section explains who talks to whom, how often, and what to do when something goes wrong. It uses the same roles shown in the OBS in Section 3, so every person can see exactly where they fit.

Messages travel in three directions: down from leadership to the teams, up from the teams to leadership, and sideways between teams that depend on each other. Two outside groups, the FYP supervisor and the site supervisor (end user), connect to the team at specific points.

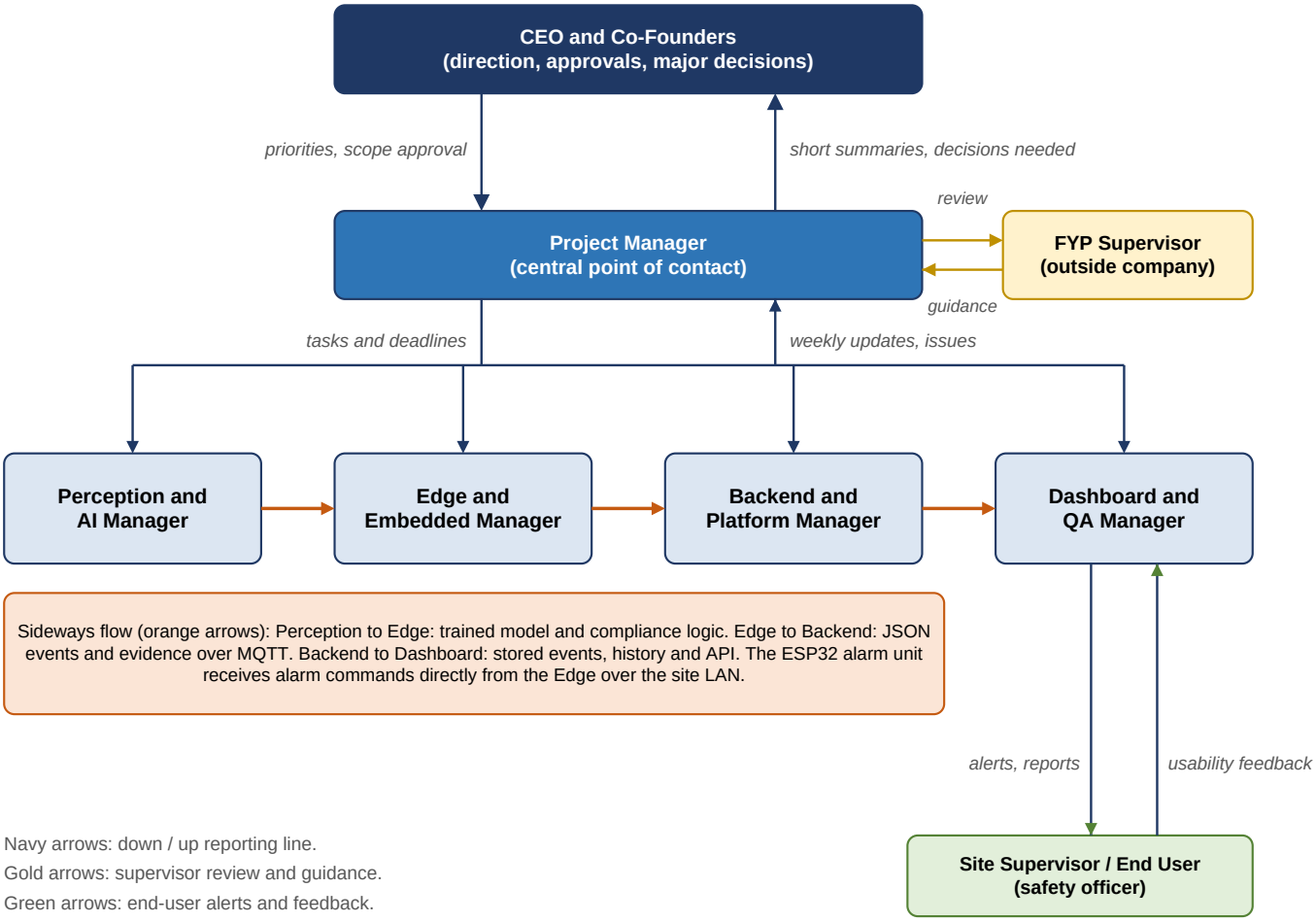

Figure 3. Communication Flow for the IntelliSafe FYP

4.2 How Information Moves

- **Downward.** The CEO and co-founders set priorities and approve the scope. The Project Manager turns these into clear tasks and deadlines for the four managers.
- **Upward.** Each manager reports progress, problems and test results to the Project Manager. The Project Manager passes a short summary to the co-founders and CEO. Anything that needs a decision travels up the same way.
- **Sideways.** The four managers talk directly to each other because their work connects. Perception hands the trained model and compliance logic to the edge side, the edge publishes JSON events and evidence over MQTT to the backend, and the backend passes stored events and history to the dashboard.
- **Outside the company.** The FYP supervisor reviews and guides the project through the Project Manager. The site supervisor gives feedback on alerts and the dashboard, mainly through the Dashboard and QA Manager.

4.3 Who Communicates with Whom

The list below follows the OBS from top to bottom. Because the two FYP students may hold more than one manager role, many of these conversations will happen between the same two people, but the same rules still apply.

CEO. Talks mainly with the three co-founders. Sets the overall direction, receives short progress summaries and makes the final call on major decisions such as scope changes or extra resources.

Co-Founders. Each co-founder follows the part of the project that matches their area. The Technology and Product co-founder handles model, edge and backend questions, the Operations and Site Safety co-founder handles site-safety requirements and site access, and the Business and Finance co-founder handles budget and equipment. They give guidance and approvals to the Project Manager and take serious issues to the

CEO.

Project Manager. The central point of contact. Passes priorities down to the managers, collects their weekly updates, keeps the task tracker and risk list, and is the link to the FYP supervisor.

Perception and AI Manager. Shares the trained model, class definitions and compliance-engine behaviour with the Edge and Embedded Manager, and agrees the JSON event format with the Backend and Platform Manager. Reports dataset or accuracy problems to the Project Manager straight away.

Edge and Embedded Manager. Runs the model and engine on the Jetson Nano, publishes events and evidence over MQTT, and drives the ESP32 alarm unit. Shares topic names, message fields and device-health information with the Backend and Platform Manager.

Backend and Platform Manager. Receives events from the edge side, checks and stores them, and shares API details and clean data with the Dashboard and QA Manager. Tells the edge side if a message format needs to change.

Dashboard and QA Manager. Shows alerts, trends and reports to the site supervisor, collects feedback, runs integration tests, and sends bug reports back to whichever team owns the problem.

FYP Supervisor. Dr. Ikramullah Khosa sits outside the company structure. Reviews progress at agreed meetings, advises on technical direction and gives feedback on the report and demonstration.

Site Supervisor / End User. Tries the prototype during demonstrations and usability testing, and tells the team what is unclear, useful or impractical on a real site.

4.4 Meetings and Updates

- **Weekly team meeting.** Everyone meets, led by the Project Manager, to go over progress, next tasks and anything blocking the work.
- **Weekly manager update.** Each manager sends a short status note to the Project Manager through the shared tracker or group chat, and again whenever a milestone is reached.
- **Technical catch-ups.** Held as needed. The managers involved agree on event formats, MQTT topics and fixes, and log the outcome in the issue tracker.
- **Supervisor reviews.** A progress meeting is held with the supervisor every two weeks, with formal reviews at weeks 1, 4, 10, 16 and 24 and the final defence at week 32. The Project Manager sends a short written update beforehand.
- **End-user feedback sessions.** Held during prototype demonstrations and usability testing, using a short feedback form.
- **Milestone approvals.** The Project Manager presents finished deliverables to the co-founders and CEO for sign-off. The team also submits key deliverables to the supervisor for academic review.

4.5 When Something Goes Wrong

Problems such as a hardware failure, lost events, a major delay or a change in scope are handled in the same order every time.

1. **Write it down.** The person who spots the problem notes what happened, what it affects and anything already tried.
2. **Tell the right people quickly.** Report it to their manager and the Project Manager by message or call, followed by a short written note.
3. **Assign an owner.** The Project Manager decides who will fix it and sets a target date.
4. **Escalate if needed.** If the issue affects scope, budget, a key milestone, data integrity or safe operation (for example, the fire/smoke alarm path failing), the Project Manager takes it to the relevant co-founder or the CEO, and to the FYP supervisor if academic guidance is needed.

5. **Record the outcome.** The decision and follow-up actions are saved in the meeting notes or project tracker.

4.6 Ground Rules

- Use one shared tracker to record each task's owner, due date, status, blockers and proof of completion.
- Write down major decisions, scope changes and supervisor feedback after every meeting.
- Fix the JSON event format up front and change it only when both the Perception and Platform leads agree, so model and platform work can proceed in parallel.
- Always include the zone or camera ID, worker_id where relevant, and timestamp in technical messages, test logs and fault reports so events can be traced.
- Raise risks early instead of waiting for the next meeting when something important is blocked.
- Share only validated test results, and clearly label measured results separately from targets.

5. Conclusion

The WBS, OBS and communication flow provide complementary views of the IntelliSafe edge safety-monitoring FYP. The WBS defines the work and expected deliverables, from dataset and model development, the compliance engine and Jetson Nano deployment to the MQTT backend, ESP32 emergency alarm, dashboard and on-device evaluation. The OBS establishes a company-style reporting model while retaining the actual FYP students and the university supervisor as project stakeholders. The communication plan explains how teams coordinate technical dependencies, report progress, receive feedback and escalate issues. Together, these structures support clearer accountability, more controlled project execution and delivery of the core system within the planned scope.